

ACID PROPERTIES & ISOLATION

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1. CREATE DATABASE

The screenshot shows the SQL Enterprise Manager interface. The left pane displays a tree view of databases, including 'bank_db', 'bank_managemen', 'bankdb', 'Tables', 'Views', 'Stored Proce', 'Functions', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', and 'practice_db'. The right pane shows the 'Acid_Properties_and_Isolation' query window with the following SQL commands:

```
1 • CREATE DATABASE Bank_ACID_DB;  
2 • USE Bank_ACID_DB;
```

Below the query window, the 'Output' tab is selected, showing the 'Action Output' table:

#	Time	Action	Message
✓ 1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
✓ 2	17:31:07	USE Bank_ACID_DB	0 row(s) affected

2. CREATE CUSTOMER TABLE

The screenshot shows the SQL Enterprise Manager interface. The left pane displays a tree view of databases, including 'bank_db', 'bank_managemen', 'bankdb', 'Tables', 'Views', 'Stored Proce', 'Functions', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', and 'students_db'. The right pane shows the 'Acid_Properties_and_Isolation' query window with the following SQL commands:

```
1 • CREATE DATABASE Bank_ACID_DB;  
2 • USE Bank_ACID_DB;  
3 • CREATE TABLE Customer (  
4     Customer_ID INT PRIMARY KEY,  
5     Customer_Name VARCHAR(100) NOT NULL,  
6     Phone VARCHAR(15),  
7     City VARCHAR(50)  
8 );
```

Below the query window, the 'Output' tab is selected, showing the 'Action Output' table:

#	Time	Action	Message
✓ 1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
✓ 2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
✓ 3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected

3. CREATE ACCOUNT TABLE

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays a tree view of database objects, including 'bank_db', 'bank_managemen', 'bankdb', 'Tables', 'Views', 'Stored Proc', 'Functions', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', 'students_db', 'sys', and 'world'. The right pane shows the 'Acid_Properties_and_Isolation' window with the following SQL script:

```
7      City VARCHAR(50)
8    );
9    CREATE TABLE Account (
10      Account_No INT PRIMARY KEY,
11      Customer_ID INT,
12      Account_Type VARCHAR(20),
13      Balance DECIMAL(12,2),
14      Branch VARCHAR(50),
15      FOREIGN KEY (Customer_ID)
16      REFERENCES Customer(Customer_ID)
17    );
18
19
```

Below the script, the 'Output' window shows the 'Action Output' table with the following data:

#	Time	Action	Message
1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected

4. CREATE TRANSACTION TABLE

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays a tree view of database objects, including 'bank_db', 'bank_managemen', 'bankdb', 'Tables', 'Views', 'Stored Proc', 'Functions', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', 'students_db', 'sys', and 'world'. The right pane shows the 'Acid_Properties_and_Isolation' window with the following SQL script:

```
15      FOREIGN KEY (Customer_ID)
16      REFERENCES Customer(Customer_ID)
17    );
18    CREATE TABLE Bank_Transaction (
19      Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
20      Account_No INT,
21      Transaction_Type VARCHAR(20),
22      Amount DECIMAL(12,2),
23      Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
24
25      FOREIGN KEY (Account_No)
26      REFERENCES Account(Account_No)
27    );
28
```

Below the script, the 'Output' window shows the 'Action Output' table with the following data:

#	Time	Action	Message
1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected

5. INSERT CUSTOMER DATA

The screenshot shows the SQL Enterprise Manager interface with the following components:

- Left Panel (Object Explorer):** Displays a tree view of database objects including 'bank_db', 'bank_managem', 'bankdb', 'Tables', 'Views', 'Stored Proce', 'Functions', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', 'students_db', 'sys', and 'world'.
- Top Panel (Query Editor):** Contains the following SQL script:


```

      23 Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
      24
      25 FOREIGN KEY (Account_No)
      26 REFERENCES Account(Account_No)
      27 );
      28 INSERT INTO Customer
      29 (Customer_ID, Customer_Name, Phone, City)
      30 VALUES
      31 (101, 'Ravi Kumar', '9876543210', 'Hyderabad'),
      32 (102, 'Priya Sharma', '9876543211', 'Vijayawada'),
      33 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
      34 (104, 'Sneha Rao', '9876543213', 'Chennai'),
      35 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
      
```
- Right Panel (SQL Additions):** Displays a message: "Automatic disabled. U manually current ca toggle a".
- Bottom Panel (Output):** Shows the execution results in a table:

#	Time	Action	Message
1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, Ph...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

6. INSERT ACCOUNT DATA

The screenshot shows the SQL Enterprise Manager interface with the following components:

- Left Panel (Object Explorer):** Displays a tree view of database objects including 'bank_db', 'bank_managem', 'bankdb', 'Tables', 'Views', 'Stored Proce', 'Functions', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', 'students_db', 'sys', and 'world'.
- Top Panel (Query Editor):** Contains the following SQL script:


```

      31 (101, 'Ravi Kumar', '9876543210', 'Hyderabad'),
      32 (102, 'Priya Sharma', '9876543211', 'Vijayawada'),
      33 (103, 'Arjun Reddy', '9876543212', 'Bangalore'),
      34 (104, 'Sneha Rao', '9876543213', 'Chennai'),
      35 (105, 'Kiran Kumar', '9876543214', 'Hyderabad');
      36 INSERT INTO Account
      37 (Account_No, Customer_ID, Account_Type, Balance, Branch)
      38 VALUES
      39 (10001, 101, 'Savings', 50000, 'Hyderabad'),
      40 (10002, 102, 'Savings', 75000, 'Vijayawada'),
      41 (10003, 103, 'Current', 120000, 'Bangalore'),
      42 (10004, 104, 'Savings', 45000, 'Chennai'),
      43 (10005, 105, 'Current', 90000, 'Hyderabad');
      
```
- Right Panel (SQL Additions):** Displays a message: "Automatic disabled. U manually current ca toggle a".
- Bottom Panel (Output):** Shows the execution results in a table:

#	Time	Action	Message
1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, Ph...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Accou...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

7. INSERT BANK TRANSACTION DATA

The screenshot shows the SQL Developer interface with the following components:

- Left Panel:** A tree view of database objects including bank_db, bank_management, bankdb, Tables, Views, Stored Procedures, Functions, bookflow_db, emr_portal, hospital_db, joins_db, practice_db, sakila, student_db, student_info, student_marks1, student_marks2, students_db, sys, and world.
- SQL Editor:** Contains SQL code for creating a database, tables, and inserting data. The current query is an INSERT statement into the Bank_Transaction table.
- Output Panel:** Displays the execution results of the SQL statements.

SQL Code:

```
39 (10001, 101, 'Savings', 50000, 'Hyderabad'),
40 (10002, 102, 'Savings', 75000, 'Vijayawada'),
41 (10003, 103, 'Current', 120000, 'Bangalore'),
42 (10004, 104, 'Savings', 45000, 'Chennai'),
43 (10005, 105, 'Current', 90000, 'Hyderabad');
44 INSERT INTO Bank_Transaction
45 (Account_No, Transaction_Type, Amount)
46 VALUES
47 (10001, 'DEPOSIT', 10000),
48 (10002, 'DEPOSIT', 15000),
49 (10003, 'WITHDRAW', 20000),
50 (10004, 'DEPOSIT', 5000),
51 (10005, 'WITHDRAW', 10000);
```

Output Panel:

#	Time	Action	Message
1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, Ph...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Accou...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transaction...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

8. DISPLAY DATA

The screenshot shows the SQL Developer interface with the following components:

- Left Panel:** Same as the previous screenshot, showing the database object tree.
- SQL Editor:** Contains SQL code for selecting data from the Customer and Account tables.
- Output Panel:** Displays the execution results, including a result grid for the SELECT queries.

SQL Code:

```
52 SELECT * FROM Customer;
53 SELECT * FROM Account;
```

Result Grid:

Customer_ID	Customer_Name	Phone	City
101	Ravi Kumar	9876543210	Hyderabad
102	Priya Sharma	9876543211	Vijayawada
103	Arjun Reddy	9876543212	Bangalore
104	Sneha Rao	9876543213	Chennai
105	Kiran Kumar	9876543214	Hyderabad

Output Panel:

#	Time	Action	Message
1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMAR...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMARY ...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, P...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Acco...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transactio...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned

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SQLAdditions

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Iter objects

- bank_db
- bank_manageme
- bankdb
- Tables
- Views
- Stored Proce
- Functions
- bookflow_db
- emr_portal
- hospital_db
- joins_db
- practice_db
- sakila
- student_db
- student_info
- student_marks1
- student_marks_c
- students_db
- sys
- world

```

48 (10002, 'DEPOSIT', 15000),
49 (10003, 'WITHDRAW', 20000),
50 (10004, 'DEPOSIT', 5000),
51 (10005, 'WITHDRAW', 10000);
52 • SELECT * FROM Customer;
53 • SELECT * FROM Account;
54 • SELECT * FROM Bank_Transaction;

```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savin Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

Account 2 x

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Output

Action Output

#	Time	Action	Message
1	17:31:07	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMAR...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMARY ...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, P...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Acco...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transactio...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned

ct Info Session

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SQLAdditions

Automa disabled. manual toggle

Iter objects

- bank_db
- bank_manageme
- bankdb
- Tables
- Views
- Stored Proce
- Functions
- bookflow_db
- emr_portal
- hospital_db
- joins_db
- practice_db
- sakila
- student_db
- student_info
- student_marks1
- student_marks_c
- students_db
- sys
- world

```

48 (10002, 'DEPOSIT', 15000),
49 (10003, 'WITHDRAW', 20000),
50 (10004, 'DEPOSIT', 5000),
51 (10005, 'WITHDRAW', 10000);
52 • SELECT * FROM Customer;
53 • SELECT * FROM Account;
54 • SELECT * FROM Bank_Transaction;

```

Result Grid

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10004	DEPOSIT	10000.00	2026-08-25 17:41:14
2	10003	WITHDRAW	15000.00	2026-08-25 17:41:14
3	10003	WITHDRAW	20000.00	2026-08-25 17:41:14
4	10004	DEPOSIT	5000.00	2026-08-25 17:41:14
5	10005	WITHDRAW	10000.00	2026-08-25 17:41:14

Bank_Transaction 3 x

Apply Revert Context Help

Output

Action Output

#	Time	Action	Message
2	17:31:07	USE Bank_ACID_DB	0 row(s) affected
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMA...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMAR...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID I...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned

ct Info Session

PART A – TRANSACTIONS

9. START A TRANSACTION

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation". The script contains the following SQL statements:

```
42 (10004, 104, 'Savings', 45000, 'Chennai'),
43 (10005, 105, 'Current', 90000, 'Hyderabad');
44 INSERT INTO Bank_Transaction
45 (Account_No, Transaction_Type, Amount)
46 VALUES
47 (10001, 'DEPOSIT', 10000),
48 (10002, 'DEPOSIT', 15000),
49 (10003, 'WITHDRAW', 20000),
50 (10004, 'DEPOSIT', 5000),
51 (10005, 'WITHDRAW', 10000);
52 SELECT * FROM Customer;
53 SELECT * FROM Account;
54 SELECT * FROM Bank_Transaction;
55 START TRANSACTION;
56
```

The Output window shows the execution results:

#	Time	Action	Message
3	17:37:02	CREATE TABLE Customer (Customer_ID INT PRIMAR...	0 row(s) affected
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMAR...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID I...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected

10. DEPOSIT MONEY

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation". The script contains the following SQL statements:

```
45 (Account_No, Transaction_Type, Amount)
46 VALUES
47 (10001, 'DEPOSIT', 10000),
48 (10002, 'DEPOSIT', 15000),
49 (10003, 'WITHDRAW', 20000),
50 (10004, 'DEPOSIT', 5000),
51 (10005, 'WITHDRAW', 10000);
52 SELECT * FROM Customer;
53 SELECT * FROM Account;
54 SELECT * FROM Bank_Transaction;
55 START TRANSACTION;
56 UPDATE Account
57 SET Balance = Balance + 5000
58 WHERE Account_No = 10001;
59
```

The Output window shows the execution results:

#	Time	Action	Message
4	17:37:49	CREATE TABLE Account (Account_No INT PRIMAR...	0 row(s) affected
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID I...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning..

11. CHECK BALANCE

The screenshot shows the SQL Enterprise Manager interface. The left pane displays a tree view of databases, including 'bank_db', 'bank_managemen', 'bankdb', 'Tables', 'Views', 'Stored Proce', 'Functions', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', 'students_db', 'sys', and 'world'. The right pane shows a query window titled 'Acid_Properties_and_Isolation*' with the following SQL code:

```

56 UPDATE Account
57 SET Balance = Balance + 5000
58 WHERE Account_No = 10001;
59 SELECT *
60 FROM Account
61 WHERE Account_No = 10001;
62

```

Below the query window, the 'Result Grid' shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

The 'Output' pane shows the 'Action Output' with the following results:

#	Time	Action	Message
5	17:38:26	CREATE TABLE Bank_Transaction (Transaction_ID I...	0 row(s) affected
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Rec
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Rec
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Rec
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Row
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned

12. COMMIT TRANSACTION

The screenshot shows the SQL Enterprise Manager interface. The left pane displays a tree view of databases, including 'bank_db', 'bank_managemen', 'bankdb', 'Tables', 'Views', 'Stored Proce', 'Functions', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', 'students_db', 'sys', and 'world'. The right pane shows a query window titled 'Acid_Properties_and_Isolation*' with the following SQL code:

```

49 (10003, 'WITHDRAW', 20000),
50 (10004, 'DEPOSIT', 5000),
51 (10005, 'WITHDRAW', 10000);
52 SELECT * FROM Customer;
53 SELECT * FROM Account;
54 SELECT * FROM Bank_Transaction;
55 START TRANSACTION;
56 UPDATE Account
57 SET Balance = Balance + 5000
58 WHERE Account_No = 10001;
59 SELECT *
60 FROM Account
61 WHERE Account_No = 10001;
62 COMMIT;
63

```

Below the query window, the 'Output' pane shows the 'Action Output' with the following results:

#	Time	Action	Message
6	17:39:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	5 row(s) affected Reco
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Reco
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Reco
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected

13. CHECK BALANCE AFTER COMMIT

The screenshot shows the SQL Enterprise Manager interface. The query editor contains the following SQL code:

```
60 FROM Account
61 WHERE Account_No = 10001;
62 COMMIT;
63 SELECT *
64 FROM Account
65 WHERE Account_No = 10001;
66
```

The Results pane displays the following table:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

The Output pane shows the execution log:

#	Time	Action	Message
7	17:40:30	INSERT INTO Account (Account_No, Customer_ID, Acc...	5 row(s) affected Recor
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Recor
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned

PART B – ROLLBACK

14. START TRANSACTION

The screenshot shows the SQL Enterprise Manager interface. The query editor contains the following SQL code:

```
53 SELECT * FROM Account;
54 SELECT * FROM Bank_Transaction;
55 START TRANSACTION;
56 UPDATE Account
57 SET Balance = Balance + 5000
58 WHERE Account_No = 10001;
59 SELECT *
60 FROM Account
61 WHERE Account_No = 10001;
62 COMMIT;
63 SELECT *
64 FROM Account
65 WHERE Account_No = 10001;
66 START TRANSACTION;
67
```

The Output pane shows the execution log:

#	Time	Action	Message
8	17:41:14	INSERT INTO Bank_Transaction (Account_No, Transact...	5 row(s) affected Re
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rc
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected

15. WITHDRAW MONEY

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation*". The script contains the following SQL statements:

```
56 • UPDATE Account
57   SET Balance = Balance + 5000
58   WHERE Account_No = 10001;
59 • SELECT *
60   FROM Account
61   WHERE Account_No = 10001;
62 • COMMIT;
63 • SELECT *
64   FROM Account
65   WHERE Account_No = 10001;
66 • START TRANSACTION;
67 • UPDATE Account
68   SET Balance = Balance - 10000
69   WHERE Account_No = 10001;
70
```

The Output window shows the execution results:

#	Time	Action	Message
9	17:43:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1

16. CHECK BALANCE

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation*". The script contains the following SQL statements:

```
66 • START TRANSACTION;
67 • UPDATE Account
68   SET Balance = Balance - 10000
69   WHERE Account_No = 10001;
70 • SELECT *
71   FROM Account
72   WHERE Account_No = 10001;
```

The Result Grid shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad

The Output window shows the execution results:

#	Time	Action	Message
10	17:43:50	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned

17. ROLLBACK

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation*". The script contains the following SQL statements:

```
60 FROM Account
61 WHERE Account_No = 10001;
62 COMMIT;
63 SELECT *
64 FROM Account
65 WHERE Account_No = 10001;
66 START TRANSACTION;
67 UPDATE Account
68 SET Balance = Balance - 10000
69 WHERE Account_No = 10001;
70 SELECT *
71 FROM Account
72 WHERE Account_No = 10001;
73 ROLLBACK;
```

The "Output" window shows the "Action Output" table with the following data:

#	Time	Action	Message
11	17:44:13	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected R
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected R
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
20	17:51:09	ROLLBACK	0 row(s) affected

18. CHECK BALANCE AFTER ROLLBACK

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation*". The script contains the following SQL statements:

```
70 SELECT *
71 FROM Account
72 WHERE Account_No = 10001;
73 ROLLBACK;
74 SELECT *
75 FROM Account
76 WHERE Account_No = 10001;
```

The "Result Grid" shows the data for the Account table:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

The "Output" window shows the "Action Output" table with the following data:

#	Time	Action	Message
12	17:45:11	START TRANSACTION	0 row(s) affected
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected F
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected F
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
20	17:51:09	ROLLBACK	0 row(s) affected
21	17:51:50	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned

PART C – SAVEPOINT

19. START TRANSACTION

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation*". The script contains the following SQL statements:

```
64 FROM Account
65 WHERE Account_No = 10001;
66 START TRANSACTION;
67 UPDATE Account
68 SET Balance = Balance - 10000
69 WHERE Account_No = 10001;
70 SELECT *
71 FROM Account
72 WHERE Account_No = 10001;
73 ROLLBACK;
74 SELECT *
75 FROM Account
76 WHERE Account_No = 10001;
77 START TRANSACTION;
78
```

The Output window shows the execution results:

#	Time	Action	Message
13	17:45:43	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected F
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected F
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
20	17:51:09	ROLLBACK	0 row(s) affected
21	17:51:50	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
22	17:52:36	START TRANSACTION	0 row(s) affected

20. FIRST UPDATE

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation*". The script contains the following SQL statements:

```
67 UPDATE Account
68 SET Balance = Balance - 10000
69 WHERE Account_No = 10001;
70 SELECT *
71 FROM Account
72 WHERE Account_No = 10001;
73 ROLLBACK;
74 SELECT *
75 FROM Account
76 WHERE Account_No = 10001;
77 START TRANSACTION;
78 UPDATE Account
79 SET Balance = Balance + 5000
80 WHERE Account_No = 10001;
81
```

The Output window shows the execution results:

#	Time	Action	Message
14	17:46:16	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning..
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
20	17:51:09	ROLLBACK	0 row(s) affected
21	17:51:50	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
22	17:52:36	START TRANSACTION	0 row(s) affected
23	17:53:07	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning..

21. CREATE SAVEPOINT

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation". The script contains the following SQL statements:

```
68 SET Balance = Balance - 10000
69 WHERE Account_No = 10001;
70 SELECT *
71 FROM Account
72 WHERE Account_No = 10001;
73 ROLLBACK;
74 SELECT *
75 FROM Account
76 WHERE Account_No = 10001;
77 START TRANSACTION;
78 UPDATE Account
79 SET Balance = Balance + 5000
80 WHERE Account_No = 10001;
81 SAVEPOINT Deposit1;
82
```

The Output window shows the execution results:

#	Time	Action	Message
15	17:46:53	COMMIT	0 row(s) affected
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning.
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
20	17:51:09	ROLLBACK	0 row(s) affected
21	17:51:50	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
22	17:52:36	START TRANSACTION	0 row(s) affected
23	17:53:07	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning.
24	17:53:40	SAVEPOINT Deposit1	0 row(s) affected

22. SECOND UPDATE

The screenshot shows the SQL Developer interface with a script titled "Acid_Properties_and_Isolation". The script contains the following SQL statements:

```
71 FROM Account
72 WHERE Account_No = 10001;
73 ROLLBACK;
74 SELECT *
75 FROM Account
76 WHERE Account_No = 10001;
77 START TRANSACTION;
78 UPDATE Account
79 SET Balance = Balance + 5000
80 WHERE Account_No = 10001;
81 SAVEPOINT Deposit1;
82 UPDATE Account
83 SET Balance = Balance - 3000
84 WHERE Account_No = 10002;
85
```

The Output window shows the execution results:

#	Time	Action	Message
16	17:47:24	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning.
19	17:49:58	SELECT * FROM Account WHERE Account_No = 10001	1 row(s) returned
20	17:51:09	WHERE Account_No = 10001 LIMIT 0, 1000	0 row(s) affected
21	17:51:50	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
22	17:52:36	START TRANSACTION	0 row(s) affected
23	17:53:07	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning.
24	17:53:40	SAVEPOINT Deposit1	0 row(s) affected
25	17:54:10	UPDATE Account SET Balance = Balance - 3000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning.

23. CREATE SECOND SAVEPOINT

The screenshot shows the SQL Enterprise Manager interface. The query editor displays a script with the following SQL statements:

```
72 WHERE Account_No = 10001;
73 ROLLBACK;
74 SELECT *
75 FROM Account
76 WHERE Account_No = 10001;
77 START TRANSACTION;
78 UPDATE Account
79 SET Balance = Balance + 5000
80 WHERE Account_No = 10001;
81 SAVEPOINT Deposit1;
82 UPDATE Account
83 SET Balance = Balance - 3000
84 WHERE Account_No = 10002;
85 SAVEPOINT Withdrawal1;
86
```

The Output pane shows the execution results:

#	Time	Action	Message
17	17:48:46	START TRANSACTION	0 row(s) affected
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Row
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
20	17:51:09	ROLLBACK	0 row(s) affected
21	17:51:50	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
22	17:52:36	START TRANSACTION	0 row(s) affected
23	17:53:07	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Row
24	17:53:40	SAVEPOINT Deposit1	0 row(s) affected
25	17:54:10	UPDATE Account SET Balance = Balance - 3000 WHE...	1 row(s) affected Row
26	17:54:38	SAVEPOINT Withdrawal1	0 row(s) affected

24. THIRD UPDATE

The screenshot shows the SQL Enterprise Manager interface. The query editor displays a script with the following SQL statements:

```
75 FROM Account
76 WHERE Account_No = 10001;
77 START TRANSACTION;
78 UPDATE Account
79 SET Balance = Balance + 5000
80 WHERE Account_No = 10001;
81 SAVEPOINT Deposit1;
82 UPDATE Account
83 SET Balance = Balance - 3000
84 WHERE Account_No = 10002;
85 SAVEPOINT Withdrawal1;
86 UPDATE Account
87 SET Balance = Balance + 10000
88 WHERE Account_No = 10003;
89
```

The Output pane shows the execution results:

#	Time	Action	Message
18	17:49:23	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
20	17:51:09	ROLLBACK	0 row(s) affected
21	17:51:50	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
22	17:52:36	START TRANSACTION	0 row(s) affected
23	17:53:07	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
24	17:53:40	SAVEPOINT Deposit1	0 row(s) affected
25	17:54:10	UPDATE Account SET Balance = Balance - 3000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...
26	17:54:38	SAVEPOINT Withdrawal1	0 row(s) affected
27	17:55:12	UPDATE Account SET Balance = Balance + 10000 WH...	1 row(s) affected Rows matched: 1 Changed: 1 Warning...

25. ROLLBACK TO SAVEPOINT

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation*". The query contains the following SQL statements:

```
76 WHERE Account_No = 10001;
77 START TRANSACTION;
78 UPDATE Account
79 SET Balance = Balance + 5000
80 WHERE Account_No = 10001;
81 SAVEPOINT Deposit1;
82 UPDATE Account
83 SET Balance = Balance - 3000
84 WHERE Account_No = 10002;
85 SAVEPOINT Withdrawal1;
86 UPDATE Account
87 SET Balance = Balance + 10000
88 WHERE Account_No = 10003;
89 ROLLBACK TO SAVEPOINT Withdrawal1;
90
```

The Output window shows the execution results:

#	Time	Action	Message
19	17:49:58	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
20	17:51:09	ROLLBACK	0 row(s) affected
21	17:51:50	SELECT * FROM Account WHERE Account_No = 1000...	1 row(s) returned
22	17:52:36	START TRANSACTION	0 row(s) affected
23	17:53:07	UPDATE Account SET Balance = Balance + 5000 WHE...	1 row(s) affected R
24	17:53:40	SAVEPOINT Deposit1	0 row(s) affected
25	17:54:10	UPDATE Account SET Balance = Balance - 3000 WHE...	1 row(s) affected R
26	17:54:38	SAVEPOINT Withdrawal1	0 row(s) affected
27	17:55:12	UPDATE Account SET Balance = Balance + 10000 WH...	1 row(s) affected R
28	17:55:43	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected

26. COMMIT

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation*". The query contains the following SQL statements:

```
78 UPDATE Account
79 SET Balance = Balance + 5000
80 WHERE Account_No = 10001;
81 SAVEPOINT Deposit1;
82 UPDATE Account
83 SET Balance = Balance - 3000
84 WHERE Account_No = 10002;
85 SAVEPOINT Withdrawal1;
86 UPDATE Account
87 SET Balance = Balance + 10000
88 WHERE Account_No = 10003;
89 ROLLBACK TO SAVEPOINT Withdrawal1;
90 COMMIT;
91
```

The Output window shows the execution results:

#	Time	Action	Message
27	17:55:12	UPDATE Account SET Balance = Balance + 10000 WH...	1 row(s) affected R
28	17:55:43	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
29	17:56:09	COMMIT	0 row(s) affected

27. CHECK ACCOUNTS

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation*". The query is as follows:

```
88 WHERE Account_No = 10003;
89 ROLLBACK TO SAVEPOINT Withdrawal1;
90 COMMIT;
91 SELECT *
92 FROM Account;
93
```

The "Result Grid" displays the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	60000.00	Hyderabad
10002	102	Savings	72000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

The "Output" pane shows the following action log:

#	Time	Action	Message
28	17:55:43	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
29	17:56:09	COMMIT	0 row(s) affected
30	17:58:16	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned

PART D – BANK TRANSFER

28. TRANSFER MONEY

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation*". The query is as follows:

```
80 WHERE Account_No = 10001;
81 SAVEPOINT Deposit1;
82 UPDATE Account
83 SET Balance = Balance - 3000
84 WHERE Account_No = 10002;
85 SAVEPOINT Withdrawal1;
86 UPDATE Account
87 SET Balance = Balance + 10000
88 WHERE Account_No = 10003;
89 ROLLBACK TO SAVEPOINT Withdrawal1;
90 COMMIT;
91 SELECT *
92 FROM Account;
93 START TRANSACTION;
```

The "Output" pane shows the following action log:

#	Time	Action	Message
29	17:56:09	COMMIT	0 row(s) affected
30	17:58:16	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
31	17:59:00	START TRANSACTION	0 row(s) affected

29. DEDUCT FROM SENDER

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation* x". The query contains the following SQL statements:

```
83 SET Balance = Balance - 3000
84 WHERE Account_No = 10002;
85 SAVEPOINT Withdrawal1;
86 UPDATE Account
87 SET Balance = Balance + 10000
88 WHERE Account_No = 10003;
89 ROLLBACK TO SAVEPOINT Withdrawal1;
90 COMMIT;
91 SELECT *
92 FROM Account;
93 START TRANSACTION;
94 UPDATE Account
95 SET Balance = Balance - 10000
96 WHERE Account_No = 10001;
```

The "Output" pane shows the execution results:

#	Time	Action	Message
30	17:58:16	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
31	17:59:00	START TRANSACTION	0 row(s) affected
32	17:59:35	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1

30. ADD TO RECEIVER

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation* x". The query contains the following SQL statements:

```
87 SET Balance = Balance + 10000
88 WHERE Account_No = 10003;
89 ROLLBACK TO SAVEPOINT Withdrawal1;
90 COMMIT;
91 SELECT *
92 FROM Account;
93 START TRANSACTION;
94 UPDATE Account
95 SET Balance = Balance - 10000
96 WHERE Account_No = 10001;
97 UPDATE Account
98 SET Balance = Balance + 10000
99 WHERE Account_No = 10002;
100
```

The "Output" pane shows the execution results:

#	Time	Action	Message
31	17:59:00	START TRANSACTION	0 row(s) affected
32	17:59:35	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1 Warr
33	18:00:05	UPDATE Account SET Balance = Balance + 10000 WH...	1 row(s) affected Rows matched: 1 Changed: 1 Warr

31. CHECK BOTH ACCOUNTS

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation". The query is as follows:

```
98 SET Balance = Balance + 10000
99 WHERE Account_No = 10002;
100 SELECT *
101 FROM Account
102 WHERE Account_No IN (10001,10002);
103
```

The "Result Grid" shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada

The "Output" window shows the following messages:

#	Time	Action	Message
32	17:59:35	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows match
33	18:00:05	UPDATE Account SET Balance = Balance + 10000 WH...	1 row(s) affected Rows match
34	18:00:31	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned

32. COMMIT TRANSFER

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation". The query is as follows:

```
91 SELECT *
92 -- Execute the selected portion of the script or everything, if there is no selection
93 START TRANSACTION;
94 UPDATE Account
95 SET Balance = Balance - 10000
96 WHERE Account_No = 10001;
97 UPDATE Account
98 SET Balance = Balance + 10000
99 WHERE Account_No = 10002;
100 SELECT *
101 FROM Account
102 WHERE Account_No IN (10001,10002);
103 COMMIT;
104
```

The "Output" window shows the following messages:

#	Time	Action	Message
33	18:00:05	UPDATE Account SET Balance = Balance + 10000 WH...	1 row(s) affected Row
34	18:00:31	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned
35	18:00:58	COMMIT	0 row(s) affected

33. FINAL BALANCES

The screenshot shows the SQL Enterprise Manager interface. The query editor contains the following SQL code:

```
104 • SELECT
105     Account_No,
106     Balance
107 FROM Account
108 WHERE Account_No IN (10001,10002);
109
```

The Results pane displays the following data:

Account_No	Balance
10001	50000.00
10002	82000.00
NULL	NULL

The Output pane shows the following messages:

#	Time	Action	Message
34	18:00:31	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned
35	18:00:58	COMMIT	0 row(s) affected
36	18:01:30	SELECT Account_No, Balance FROM Account WH...	2 row(s) returned

PART E – TRANSFER USING ROLLBACK

34. START TRANSACTION

The screenshot shows the SQL Enterprise Manager interface. The query editor contains the following SQL code:

```
96 WHERE Account_No = 10001;
97 • UPDATE Account
98 SET Balance = Balance + 10000
99 WHERE Account_No = 10002;
100 • SELECT *
101 FROM Account
102 WHERE Account_No IN (10001,10002);
103 • COMMIT;
104 • SELECT
105     Account_No,
106     Balance
107 FROM Account
108 WHERE Account_No IN (10001,10002);
109 • START TRANSACTION;
```

The Output pane shows the following messages:

#	Time	Action	Message
35	18:00:58	COMMIT	0 row(s) affected
36	18:01:30	SELECT Account_No, Balance FROM Account WH...	2 row(s) returned
37	18:02:07	START TRANSACTION	0 row(s) affected

35. DEDUCT MONEY

The screenshot shows the SQL Server Enterprise Manager interface. The query window displays the following SQL code:

```
100 • SELECT *
101 FROM Account
102 WHERE Account_No IN (10001,10002);
103 • COMMIT;
104 • SELECT
105     Account_No,
106     Balance
107 FROM Account
108 WHERE Account_No IN (10001,10002);
109 • START TRANSACTION;
110 • UPDATE Account
111     SET Balance = Balance - 20000
112     WHERE Account_No = 10001;
113
```

The Output pane shows the execution results:

#	Time	Action	Message
✓ 36	18:01:30	SELECT Account_No, Balance FROM Account WH...	2 row(s) returned
✓ 37	18:02:07	START TRANSACTION	0 row(s) affected
✓ 38	18:02:33	UPDATE Account SET Balance = Balance - 20000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1

36. ADD MONEY

The screenshot shows the SQL Server Enterprise Manager interface. The query window displays the following SQL code:

```
103 • COMMIT;
104 • SELECT
105     Account_No,
106     Balance
107 FROM Account
108 WHERE Account_No IN (10001,10002);
109 • START TRANSACTION;
110 • UPDATE Account
111     SET Balance = Balance - 20000
112     WHERE Account_No = 10001;
113 • UPDATE Account
114     SET Balance = Balance + 20000
115     WHERE Account_No = 10002;
116
```

The Output pane shows the execution results:

#	Time	Action	Message
✓ 37	18:02:07	START TRANSACTION	0 row(s) affected
✓ 38	18:02:33	UPDATE Account SET Balance = Balance - 20000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1
✓ 39	18:03:04	UPDATE Account SET Balance = Balance + 20000 WH...	1 row(s) affected Rows matched: 1 Changed: 1

37. CANCEL TRANSACTION

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation*". The query contains the following SQL statements:

```
104 • SELECT
105     Account_No,
106     Balance
107 FROM Account
108 WHERE Account_No IN (10001,10002);
109 • START TRANSACTION;
110 • UPDATE Account
111     SET Balance = Balance - 20000
112     WHERE Account_No = 10001;
113 • UPDATE Account
114     SET Balance = Balance + 20000
115     WHERE Account_No = 10002;
116 • ROLLBACK;
```

The "Output" pane shows the "Action Output" table with the following data:

#	Time	Action	Message
38	18:02:33	UPDATE Account SET Balance = Balance - 20000 WHE...	1 row(s) affected R
39	18:03:04	UPDATE Account SET Balance = Balance + 20000 WH...	1 row(s) affected R
40	18:03:29	ROLLBACK	0 row(s) affected

38. CHECK BALANCES

The screenshot shows the SQL Developer interface with a query window titled "Acid_Properties_and_Isolation*". The query contains the following SQL statements:

```
115 WHERE Account_No = 10002;
116 • ROLLBACK;
117 • SELECT *
118 FROM Account
119 WHERE Account_No IN (10001,10002);
120
```

The "Result Grid" shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada

The "Output" pane shows the "Action Output" table with the following data:

#	Time	Action	Message
39	18:03:04	UPDATE Account SET Balance = Balance + 20000 WH...	1 row(s) affected R
40	18:03:29	ROLLBACK	0 row(s) affected
41	18:03:58	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned

PART F – ACID

39. ATOMICITY

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Edit View Query Database Server Tools Scripting Help

SQLAd

Acid_Properties_and_Isolation*

Limit to 1000 rows

```
120 • START TRANSACTION;
121 • UPDATE Account
122   SET Balance = Balance - 10000
123   WHERE Account_No = 10001;
124 • UPDATE Account
125   SET Balance = Balance + 10000
126   WHERE Account_No = 10002;
127 • COMMIT;
```

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	82000.00	Vijayawada
NULL	NULL	NULL	NULL	NULL

Result Grid

Form Editor

Account 12

Apply Revert Context

Output

Action Output

#	Time	Action	Message
✓ 42	18:06:00	START TRANSACTION	0 row(s) affected
✓ 43	18:06:00	UPDATE Account SET Balance = Balance - 20000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1
✓ 44	18:06:00	UPDATE Account SET Balance = Balance + 20000 WH...	1 row(s) affected Rows matched: 1 Changed: 1
✓ 45	18:06:00	ROLLBACK	0 row(s) affected
✓ 46	18:06:00	SELECT * FROM Account WHERE Account_No IN (100...	2 row(s) returned
✓ 47	18:06:00	START TRANSACTION	0 row(s) affected
✓ 48	18:06:00	UPDATE Account SET Balance = Balance - 10000 WHE...	1 row(s) affected Rows matched: 1 Changed: 1
✓ 49	18:06:00	UPDATE Account SET Balance = Balance + 10000 WH...	1 row(s) affected Rows matched: 1 Changed: 1
✓ 50	18:06:00	COMMIT	0 row(s) affected
✓ 51	18:06:00	ROLLBACK	0 row(s) affected

Session